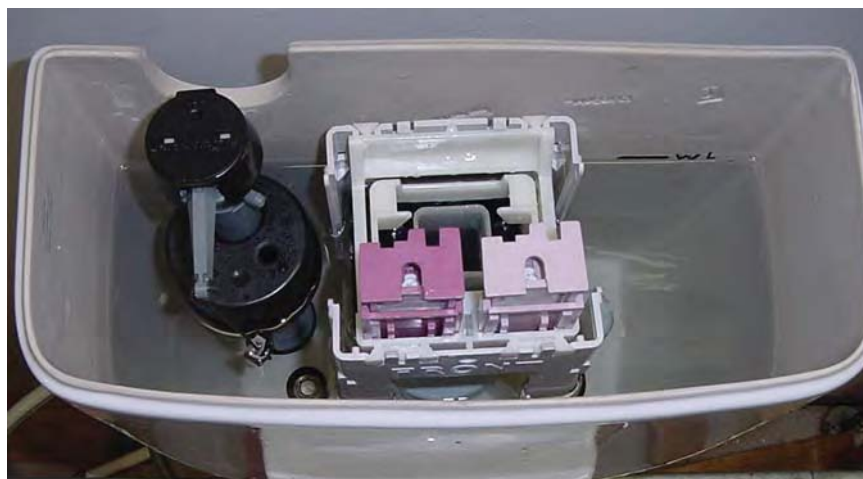


The Real High-Efficiency Toilets Have Arrived

by John Koeller

Why discuss toilets in an energy publication? The interesting secret about toilets—and all of the other fixtures and appliances that use water—is that they are responsible for a good part of a region's energy consumption. Why? Because pumping water, delivering water, and treating wastewater all require electricity. About 5% of all electricity consumed in California is used to provide these essential water-related services. Another 14% is used on the customer side of the meter to water lawns, take showers, and so forth. In short, 19% of all electricity used statewide is used for water-related activities. And according to various field studies, toilet flushing in the typical home is responsible for 27% of indoor water use. Add to that the leakage that is found with older fixtures, and the figure can rise to well above 30%—more than for any other indoor use! Clearly, improving the performance and efficiency of toilets will save energy.

From a consumer perspective, toilets spark either strong dissatisfaction or quiet acceptance. Seldom will you find people raving with pleasure over their toilets. Thus, when new ideas and technologies appear that seem to threaten the performance of our thrones, personal antennae go up and we may exhibit strong prejudices. Such has been the case with the new high-efficiency toilets now available in the marketplace. But before I discuss these refined and redesigned fixtures, let's



VERITEC CONSULTING, INC.

This dual-flush toilet offers the user a choice of either a 1.6-gallon flush or a 0.8-gallon flush.

take a quick look at the recent history of toilets in North America.

THE EARLY DAYS

Before the 1950s, the typical residential toilet flushed with 7 gallons of water. From the '50s to the mid-'70s, the trend was downward toward 5 gallons per flush. In the late 1970s, toilet manufacturers introduced the water-saver fixture, which used only 3.5 gallons. Through the 1980s, 3.5 gallons per flush remained the standard.

By the late 1980s, however, advocates of water efficiency noted that some European toilets were functioning at 1.6 gallons per flush, or less. These advocates began to promote the use of such toilets here. Between 1989 and 1993, 16 states and six localities adopted standards that mandated a maximum of 1.6 gallons per flush. The ultralow-flush toilet was born.

DRAMATIC CHANGES IN THE MARKETPLACE

By 1992, all major toilet manufacturers were offering gravity-fed models that functioned at 1.6 gallons per flush. In the same year, the plumbing industry saw chaos overtaking the marketplace; it looked as if each of the 50 states was going to enact a different set of standards for toilets. They joined with water and wastewater utilities and environmental organizations to call upon Congress to enact nationwide standards that would preempt all of the various state and local standards. The result was the Energy Policy Act of 1992, which established a maximum of 1.6 gallons per flush for all tank-type toilets manufactured or imported after January 1, 1994.

Did that solve the problem? It did not. Some advocates of the 1.6-gallon toilet had based their enthusiasm and their claims of efficiency and

performance on European designs. Those designs had sold well in Europe for some time, and after all, they reasoned, what worked in Europe would work here—right? Well, not exactly. It seems no one in government stopped to consider that European toilets work quite differently from the 1.6-gallon toilets

decided to redesign their 3.5-gallon siphonic models so that they would work properly with only 1.6 gallons of water. Some manufacturers were able to accomplish this much more speedily and effectively than others.

It was a serious challenge, and in the end, the new water-efficient toilets often presented two problems.

had to deal with the negative backlash caused by the inferior models that other manufacturers were selling.

PROBLEMS CORRECTED

Fortunately, and to the credit of the plumbing industry, the manufacturers acknowledged these early problems and corrected them. By the late 1990s, most manufacturers had abandoned those early designs and were bringing their second-generation models to market. Improved performance yielded satisfied customers. The customer satisfaction surveys conducted by the water utility industry clearly proved that.

Late 1998 saw the introduction of the third-generation fixture—the high-efficiency toilet, which functioned at 1.28 gallons per flush—a flush volume 20% below that of a conventional ultralow-flush toilet.

The first high-efficiency toilet to arrive in the North American marketplace was a gravity-fed dual-flush design, which had been available in Australia for over ten years. A dual-flush toilet offers the user two flush options: a full (1.6-gallon) option and a reduced (0.8–1.1-gallon) option for liquids only (see photo, p. 10). Users choose the appropriate option each time they flush.

Field studies in the United States and Canada showed that, in residential applications, the weighted average real-world flush volume of dual-flush fixtures was approximately 1.25 gallons per flush, which qualified them as high-efficiency toilets. Today, 16 manufacturers offer 71 different models of dual-flush high-efficiency toilets, all of them suited to residential installation. Of these, the vast majority are gravity fed, while the rest are pressure assist models.

The second high-efficiency toilet to appear in the marketplace was the single-flush pressure assist design (see photo, this page). Water authorities in California conducted field trials of the first prototypes of these fixtures



VERTIC CONSULTING, INC.

This pressure assist single-flush toilet uses only 1 gallon per flush.

that were being installed in North America. European toilets have a wash down bowl design, while the North American toilets had a siphonic bowl design. We could live with that design difference—but we couldn't get past the fact that European toilets do not physically fit our plumbing systems. European toilets are usually designed to fit a 5-inch rough-in, while the rough-in dimensions for most North American toilets are 10 to 14 inches. (The rough-in is the horizontal distance from the wall behind the toilet to the centerline of the toilet floor flange.) It wasn't going to be possible to transfer European technology directly to North America.

To make toilets that people would buy, manufacturers needed to develop new designs for the North American market. Yet they couldn't fast-track the development of new bowl designs by basing them directly on European designs. Instead, they

The first was that they did not always flush adequately. The second was that flush volume was inconsistent and sometimes greater than the permitted maximum of 1.6 gallons. Customers were dissatisfied with the poor performance, while water-efficiency advocates were unhappy with the erratic flush volume.

Now it is an old saying that if everyone on both sides of your particular issue is unhappy with you, you must be doing something right. Such was not the case, however, in the early days of 1.6-gallon toilets. Some homeowners were having to flush three or four times or continually unclog their new toilets; water agencies were not saving as much water as they had expected—or, in some instances, as they were paying for, in the case of rebated or subsidized toilets. Manufacturers were hearing complaints; and the manufacturers that were producing the best models

in 2001. By 2003, a variety of models were entering the North American marketplace. These fixtures function at 0.9–1.1 gallons per flush and have proven to be excellent performers both in laboratory testing and, more importantly, in the real world. Since only one flush volume is available, the user is not required to make a choice, which eliminates the behavioral aspect of the dual flush. Today, 11 manufacturers offer 33 different models of this technology.

Some critics argue that pressure assist toilets are too noisy for residential installations. However, the newest ones are much quieter than the pressure assist models from the 1990s. In fact, some gravity-fed toilets are noisier than today's pressure assist high-efficiency toilets. Over 4,000 of these toilets were installed in homes in Redwood City, California, over the past several years—which proves that customers like them. In the city's toilet replacement program, customers are given a choice between a conventional gravity-fed 1.6-gallon ultralow-flush toilet and a pressure assist single-flush 1-gallon high-efficiency toilet. The majority of residents have chosen the latter, fully aware that it might be somewhat noisier than the ultralow-flush toilet.

The third category of high-efficiency toilet is the gravity-fed single-flush design, which functions at 1.28 gallons per flush or below (see photo, this page). Like the pressure assist toilet, this toilet has only one flush option. Today, six different models are available from four manufacturers, but many more are expected to enter the marketplace in 2007.

Over 15,000 high-efficiency toilets have already been installed in California homes. About half of them are dual-flush models and half are pressure assist single-flush models. Within the next 12 months, we expect that the number of high-efficiency toilets available in the marketplace will grow by as much as 50%, as some manufacturers introduce many new

models and others begin to focus less on ultralow-flush toilets and more on high-efficiency toilets.

FLUSH PERFORMANCE

Remarkably, these third-generation high-efficiency toilets are a vast improvement over the excellent 1.6-gallon toilets that were sold from the late 1990s right up to today. The plumbing industry must be



This gravity-fed single-flush toilet uses 1.28 gallons per flush.

commended for the improvements in flush performance that it has made over the last eight years.

While marginal performers may still occasionally be found in the marketplace, consumers and contractors alike can now choose toilets with confidence, since authoritative information on performance—the MaP testing results—is now available. In 2003, water utilities in the United States and Canada jointly developed a set of maximum performance (MaP) testing protocols for toilet fixtures. MaP testing uses extruded soy bean paste and toilet paper as the testing

media that most closely simulate the real world demands on a toilet. The test measures the maximum amount of paste, measured in grams, that can be evacuated from a toilet in a single flush. Today, MaP testing is widely used to measure and rank flush performance of tank-type toilets for both residential and commercial applications (see photo, p. 13). (See “Which Toilets Deliver?” *HE* May/June '05, p. 19, for a full discussion of MaP testing.) The California Urban Water Conservation Council (CUWCC) publishes the results of these performance tests on the organization's Web site. For the latest (ninth edition) MaP report, which describes the performance of 332 different toilet models, go to www.cuwcc.org/maptesting.lasso.

From a performance standpoint, most high-efficiency toilets earn higher MaP scores than the average 1.6 gallons per flush ultralow-flush toilet, probably because of the extensive development and engineering that went into these third-generation models. Specifically, while new ultralow-flush toilets average 450 grams in the MaP testing protocol, the new high-efficiency toilets average 550 grams. In short, the newer high-efficiency toilets, using less water, remove more waste than the older 1.6-gallon ultralow-flush toilets, thanks to excellent reengineering on the part of the manufacturers.

When choosing fixtures, it is important to keep in mind that the MaP test represents only one measure of flush performance—that of evacuating sinking solid waste in a single flush. Noise, splashing, thoroughness of bowl cleaning, and other aspects of performance are not measured in the MaP testing protocol. These things may be just as important to customers as the amount of solid waste evacuated. It is possible (although I have no evidence to prove it) that because MaP testing is such a successful marketing tool, manufacturers may sometimes sacrifice

other aspects of performance for the sake of achieving a higher MaP score. If that is so, a toilet that scores 1,000 grams on the MaP test might not satisfy any more customers than a toilet that scores, say, 600 grams.

DRAIN LINE CONCERNS

When flush volume is reduced to 1 gallon, will there be enough water to move the waste down the drain line? Or will the drain clog and back up? To find out, Veritec Consulting, Incorporated, and Koeller and Company conducted a very comprehensive laboratory study, using the realistic MaP testing media, in 2005. They studied nine different flush technologies to learn how the solid waste moved in drain lines of different diameters and slopes. The results of the study showed that, for typical residential installations, there was sufficient water to move the waste to the sewer in all cases. More recently, Veritec Consulting conducted another lab test of high-efficiency toilets for the U. S. Environmental Protection Agency (EPA). The results of this second test were similar; they too showed that high-efficiency toilets provide sufficient water to transport solid waste.

Keep in mind that, while high-efficiency toilets must meet the same standards for drain line waste transport as all other toilets sold in the United States and Canada, the fact that a toilet meets these standards does not guarantee that the drain line will never clog. All toilets, regardless of flush volume, may drain poorly if the drain line is damaged or degraded—if it is invaded by roots, or if the lines are broken, for example—or in a building with a very long drain line run and no additional source of wastewater near the toilet.

WATERSENSE SUPPORTS HIGH-EFFICIENCY TOILETS

In early 2004, EPA began developing the WaterSense program, which is to water conservation what the highly successful Energy Star program is to energy. At first, toilets were not going to be included in the program, because there was so much controversy surrounding

This specification included MaP testing with a 350-gram minimum performance threshold. The draft was presented to stakeholders for comment. The final specification was released in January 2007 and as of this writing, product testing and labeling has begun. By including high-efficiency toilets in the WaterSense program, EPA has officially recognized the viability of this technology.



MaP testing is conducted in the test rig pictured above, using extruded soy bean paste and toilet paper as the testing media.

their performance. But with the groundwork on MaP testing and product evaluation already in place, it became apparent that toilets, which offer perhaps the greatest potential for indoor water savings, would be prime candidates for the WaterSense program. In the end, EPA not only included toilets in WaterSense, but also used toilets as one of the headlines that kicked off the start of the program.

One of EPA's first decisions was to label only high-efficiency toilets. These were defined based on the definition developed earlier by the water utilities—toilets with an effective flush volume of no more than 1.28 gallons—which EPA rounded up to 1.3 gallons per flush.

In 2006, the EPA rolled out its draft specification for high-efficiency toilets.

SHRINKING PRICE PREMIUMS

Although the CUWCC does not track fixture pricing at the retail level, anecdotal evidence suggests that dual-flush high-efficiency toilets cost about \$100 more than conventional 1.6-gallon single-flush ultralow-flush toilets. For the consumer, the simple payback of this extra cost ranges between one and two years, depending on local utility charges for water and wastewater treatment. On the other hand, there is generally little difference in price between high-

efficiency and ultralow-flush pressure assist toilets, since the technology is the same for both; It is only the internal containment vessel and the bowl design that differ.

On the wholesale level, dual-flush toilets and pressure assist single-flush toilets are priced about the same—typically from \$120 to \$145 per fixture when buying direct in truckload quantities, although pricing below \$100 per fixture is being achieved in some cases.

As demand increases, competition is intensifying among the 22 manufacturers of high-efficiency toilets. Indications are that prices are continuing to drop, as they did all through 2005 and 2006. Expect prices to soften a bit further in 2007 as home building slows down, as manufacturers

move more production to regions with lower labor costs, and as more high-efficiency toilets are produced, bringing added economies of scale.

CHOOSING A HIGH-EFFICIENCY TOILET

Although I make it a policy to not recommend specific models, here are some things for consumers, architects, specifiers, and contractors to consider when choosing a high-efficiency toilet.

- Gravity-fed models fitted with 3-inch-diameter flappers or flush valves tend to outperform models fitted with the more traditional 2-inch flappers.
- Pressure assist models tend to perform well and to deliver consistent water savings over the long run though they are usually a little louder and sometimes more expensive than gravity-fed models.
- Performance and price do not go hand-in-hand. Some inexpensive models perform very well. Conversely, some costly models perform very poorly.
- Rely on MaP test results for flush performance. Ignore the less meaningful claims that you might hear at a trade show from sales personnel. Statements such as “This model can flush 30 golf balls” or “This model has a fully glazed trapway” are usually little more than marketing gimmicks.
- Consider specific models rather than specific manufacturers. Many manufacturers offer both high-performing and low-performing models.



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For more information:

For an up-to-date listing of all of the high-efficiency toilets offered in the North American marketplace, go to www.cuwcc.org/toilet_fixtures/HET.pdf.

If you have additional questions or comments about high-efficiency toilets, MaP testing, or water utility incentive programs for high-efficiency toilets, contact

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To learn more about Redwood City's toilet replacement program, visit www.redwoodcity.org/publicworks/water/toiletprogramindex.html.

To learn more about the WaterSense program, see “Efficiency Just Makes WaterSense,” p. 4.

To download the final specification for high-efficiency toilets, go to www.epa.gov/OW-OWM.html/water-efficiency/docs/spec_het508.pdf.

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California's Water-Energy Relationship, CEC-700-2005-011-SF. November 2005. To download this report, go to www.energy.ca.gov/2005publications.

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